

Table 1: Clustering inertia on dataset collection 2 with 10.0% constraints

| Constraints Level<br>Method<br>Dataset | 0.1     |       |        |           |             |         |       |       |
|----------------------------------------|---------|-------|--------|-----------|-------------|---------|-------|-------|
|                                        | COPKM_P | KSKM  | KSKM_E | KSKM_repo | KSKM_E_repo | BLPKMCC | PCCC  | COPKM |
| synthetic_n300_d2_k10_s10              | 6.48    | 4.94  | 4.94   | 4.94      | 4.94        | 15.22   | 6.19  | 8.34  |
| synthetic_n300_d2_k10_s20              | 21.79   | 18.14 | 18.14  | 18.17     | 18.16       | 24.33   | 19.34 | 25.21 |
| synthetic_n300_d2_k10_s30              | 44.71   | 36.22 | 36.22  | 36.23     | 36.23       | 42.10   | 36.26 | 40.07 |
| synthetic_n300_d2_k10_s40              | 62.85   | 55.33 | 55.46  | 55.74     | 55.63       | 58.59   | 55.66 | 61.81 |
| synthetic_n300_d2_k10_s50              | 77.31   | 71.45 | 71.17  | 71.78     | 71.41       | 72.96   | 71.64 | 79.20 |
| synthetic_n300_d2_k20_s10              | 7.49    | 6.02  | 5.99   | 6.04      | 5.99        | 10.76   | 6.01  | 7.42  |
| synthetic_n300_d2_k20_s20              | 19.15   | 17.52 | 17.53  | 17.88     | 17.57       | 20.60   | 18.06 | 19.57 |
| synthetic_n300_d2_k20_s30              | 31.36   | 29.24 | 29.31  | 29.86     | 29.56       | 32.92   | 29.92 | 31.59 |
| synthetic_n300_d2_k20_s40              | 40.81   | 37.34 | 37.15  | 38.41     | 37.57       | 41.33   | 38.41 | 40.75 |
| synthetic_n300_d2_k20_s50              | 47.24   | 42.38 | 42.45  | 43.28     | 42.84       | 47.41   | 44.48 | 46.49 |
| synthetic_n300_d2_k50_s10              | 3.47    | 3.23  | 3.30   | 3.31      | 3.30        | 5.06    | 3.31  | 3.49  |
| synthetic_n300_d2_k50_s20              | 8.06    | 7.45  | 7.40   | 7.66      | 7.73        | 9.32    | 7.85  | 8.16  |
| synthetic_n300_d2_k50_s30              | 10.40   | 9.42  | 9.44   | 9.89      | 9.96        | 11.67   | 9.89  | 10.40 |
| synthetic_n300_d2_k50_s40              | 11.82   | 10.43 | 10.46  | 11.08     | 11.14       | 13.57   | 11.18 | 11.73 |
| synthetic_n300_d2_k50_s50              | 12.98   | 12.01 | 11.99  | 12.79     | 12.62       | 14.97   | 12.73 | 13.03 |